

Alessio Salvatore

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Education

ETH Zürich, Zürich, Switzerland Sep 2025–Present
M.Sc. in Robotics

Coursework: Probabilistic Artificial Intelligence, Advanced Machine Learning, Image Analysis and Computer Vision, Planning and Decision Making for Autonomous Robots, 3D Vision, Perception and Learning for Robotics, Deep Generative Models, Robot Learning

Politecnico di Milano, Milan, Italy Sep 2022–Jul 2025
B.Sc. in Automation Engineering

Final Grade: 110/110 *cum laude* (equivalent GPA $\approx$ 4.0/4.0; 29.57/30).

Illinois Institute of Technology, Chicago, IL, USA Aug–Dec 2024
Exchange — Global E3 Program

Projects

Research Projects

Goal-Driven Adaptive-Resolution 3D Mapping with RL  Report Feb–Jun 2026
Computer Vision and Geometry Group – Marc Pollefeys Lab

- Trained a PPO policy that adapts per-region voxel resolution (split/keep/merge) online during navigation under a hard memory budget, reformulating an intractable per-block action space into a compact two-threshold policy
- Re-implemented PEANUT's dense voxel map as a sparse hierarchical multi-resolution grid (Habitat, HM3D); the learned policy retains 79% of point-goal navigation success with 87% less memory than a uniformly fine map (20 unseen episodes)

Knowledge Distillation for Aerial Robots Control  Video  Report Feb–Jun 2026
Autonomous Systems Lab – Roland Siegwart Lab

- Selected among 150 applicants for a closed cohort of 30 students in the Perception and Learning for Robotics class at ETH Zurich
- Distilled parameter-matched model-based controllers for an overactuated tilt-rotor MAV (OMAV) into a single recurrent LSTM policy, trained closed-loop through differentiable vehicle dynamics with DAgger on ~ 42 h of simulated flight
- In simulation the policy outperforms the nominal model-based controller in velocity tracking ($p < 10^{-10}$, 2,000 paired rollouts) and nearly halves divergence under $\pm 30\%$ model mismatch; deployed zero-shot on the real OMAV

Selected Projects

Autonomous Exploration on a Unitree A2 Quadruped  Repo 06/2026
ETH Robotics Summer School (ETHZ-RobotX)

- Deployed a full autonomous search-and-explore mission with our team on a real Unitree A2 quadruped (Hesai LiDAR; TARE/FAR planners, DLIO LiDAR-inertial odometry) across three field sites
- Developed the artifact-mapping node fusing YOLO detections with TF2 transforms into a deduplicated map-frame object log – the mission's scored output

Vision-Language-Action Policies on a Real SO-101 Arm  Repo  Video 05/2026
Project of the Class Robot Learning

- Fine-tuned SmoVLVA policies for language-conditioned pick-and-place across three tasks, up to grounding celebrity portraits with generalization to identities never seen in training; on-robot inference at 10 FPS
- Built the LeRobot data pipeline (prompt relabeling, brightness and face-swap augmentation, DAgger-style corrections) and automated H100 cloud training with checkpoint streaming to the Hugging Face Hub

Person Following on the Agibot X2 Humanoid  Repo  Video 05/2026
RoboHack Hackathon

- Built real-time person following running fully onboard in a hackathon weekend: YOLOv8 on the head stereo cameras with disparity-based depth, Ruckig head tracking, and a state-machine locomotion supervisor holding a set distance

Channel Position Estimation from Pipe-Inspection Video Repo Video Slides

05/2026

Physical AI Hackathon

- Estimated per-frame absolute position, turning point, and direction of a sewer-inspection probe from video alone, via sparse optical flow with outlier filtering, scene gating, and scale-free normalization – CPU-only, ~1000 frames in under 8 s

DINOv3 for Mitral Valve Segmentation Repo Video

12/2025

- Built a unified preprocessing pipeline for heterogeneous medical video datasets, generated pseudo-labels from sparse annotations via DINO-based feature propagation, and trained a U-Net segmentation model on the filtered labels

Multi-Agent Multi-Goal Collection Planner Video

12/2025

Project of the Class Planning and Decision Making for Autonomous Robots

- Designed a multi-robot planning framework combining Large Neighborhood Search for task allocation with real-time path planning and decentralized collision avoidance under partial observability.

Autonomous Satellite Docking Planner Video

11/2025 – 12/2025

Project of the Class Planning and Decision Making for Autonomous Robots

- Implemented a trajectory optimization pipeline for autonomous satellite docking, solving non-convex guidance problems via Sequential Convexification in cluttered environments.

Work Experience

JOINT Lab — Italian Institute of Technology, Bergamo

Feb–Jun 2025

Research Intern — Industrial Robotics Repo

- CAD design of a linear actuator built from commercial off-the-shelf components.
- Implemented real-time control algorithms using Arduino.
- Developed a MATLAB simulation to validate system dynamics and behavior.

ETH Robotics Club

Nov 2025–Present

Partnerships Team Lead

- Secured sponsorships for club initiatives and events

Laelaps AI

Jun 2026–Present

Robotics Engineer

- Developing multi-agent systems for robot orchestration in complex real-world environments